Interest Rate and Credit Risk Models
Fall 2025

Project 2

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December 3, 2025

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1 Exercise 1

The Gaussian HJM model is

$$df(t,T) = \alpha(t,T)\,dt + \sigma_1\,dW_1^*(t) + \sigma_2 e^{-a(T-t)} dW_2^*(t),$$

so the forward-rate volatility vector is

$$\sigma(t,T) = \big(\sigma_1,\ \sigma_2 e^{-a(T-t)}\big)^\top.$$

(a) HJM drift. The no-arbitrage HJM drift condition follows from the fact that a discounted zero-coupon bond must be a martingale under $\mathbb{Q}$:

$$\alpha(t,T) = \sum_{k=1}^{2} \sigma_k(t,T) \int_t^T \sigma_k(t,s)\,ds.$$

Here

$$\int_t^T \sigma_1\,ds = \sigma_1(T-t), \qquad \int_t^T \sigma_2 e^{-a(s-t)} ds = \sigma_2 \frac{1-e^{-a(T-t)}}{a},$$

which yields

$$\boxed{\alpha(t,T) = \sigma_1^2(T-t) + \sigma_2^2 e^{-a(T-t)} \frac{1-e^{-a(T-t)}}{a}.}$$

In the HJM framework, once the volatilities $\sigma_k(t,T)$ are specified, the drift $\alpha(t,T)$ is completely determined; there is no additional freedom if one wants to avoid arbitrage.

(b) Instantaneous volatility and bond price dynamics. The general HJM formula is (see Lemma 23.1)

$$\frac{dP(t,T)}{P(t,T)} = \Big(r(t) - \int_t^T \alpha(t,u)\,du + \tfrac{1}{2}\|v(t,T)\|^2\Big)\,dt + v(t,T)^\top dW^*(t). \tag{1}$$

$$v(t,T) = -\int_t^T \sigma(t,u)\,du, \qquad \sigma(t,T) = (\sigma_1,\ \sigma_2 e^{-a(T-t)})^\top.$$

and the short rate

$$r(t) = f(t,t)$$

Hence, in our model,

$$v(t,T) = -\begin{pmatrix} \sigma_1(T-t) \\ \sigma_2 \dfrac{1-e^{-a(T-t)}}{a} \end{pmatrix}.$$

$$\boxed{\|v(t,T)\| = \sqrt{\sigma_1^2(T-t)^2 + \sigma_2^2 \left(\frac{1-e^{-a(T-t)}}{a}\right)^2}.}$$

By definition of the risk–neutral measure, the discounted bond price

$$\frac{P(t,T)}{B(t)}, \qquad B(t) = \exp\Big(\int_0^t r(s)\,ds\Big),$$

must be a martingale. Since $dB(t)/B(t) = r(t)\,dt$, we have

$$\frac{d\big(P(t,T)/B(t)\big)}{P(t,T)/B(t)} = \frac{dP(t,T)}{P(t,T)} - r(t)\,dt.$$

Injecting (1), we obtain

$$\frac{d\big(P(t,T)/B(t)\big)}{P(t,T)/B(t)} = \Big(-\int_t^T \alpha(t,u)\,du + \tfrac{1}{2}\|v(t,T)\|^2\Big)\,dt + v(t,T)^\top dW^*(t).$$

A continuous martingale has zero drift, so the term multiplying dt must vanish:

$$-\int_t^T \alpha(t,u)\,du + \tfrac{1}{2}\|v(t,T)\|^2 = 0. \tag{2}$$

This result can also be recovered by computing explicitly the integral of the drift term $\alpha(t,u)$ obtained in part (a) and verifying that it satisfies the required identity.
Substituting it back into (1) yields the risk–neutral bond dynamics

$$\boxed{\frac{dP(t,T)}{P(t,T)} = r(t)\,dt + v(t,T)^\top dW^*(t).}$$

(c) Numerical values and plots. For $a = 0.4$, $\sigma_1 = 0.01$, $\sigma_2 = 0.02$, $t = 0.5$ and $T \in \{0.5, 1, 2, 3\}$, one evaluates

$$\alpha(t,T) = \sigma_1^2(T-t) + \sigma_2^2 e^{-a(T-t)}\frac{1-e^{-a(T-t)}}{a}, \qquad v_1(t,T) = \sigma_1(T-t), \qquad v_2(t,T) = \sigma_2\frac{1-e^{-a(T-t)}}{a}.$$

T	$\tau = T-t$	$\alpha(t,T)$	$v_1(t,T)$	$v_2(t,T)$	$\|v(t,T)\|$
0.5	0.0	0.00000	0.00000	0.00000	0.00000
1.0	0.5	0.00020	0.00500	0.00906	0.01035
2.0	1.5	0.00040	0.01500	0.02256	0.02709
3.0	2.5	0.00048	0.02500	0.03161	0.04030

Table 1: Numerical values of $\alpha(t,T)$ and $v(t,T)$ for $a = 0.4$, $\sigma_1 = 0.01$, $\sigma_2 = 0.02$, $t = 0.5$.

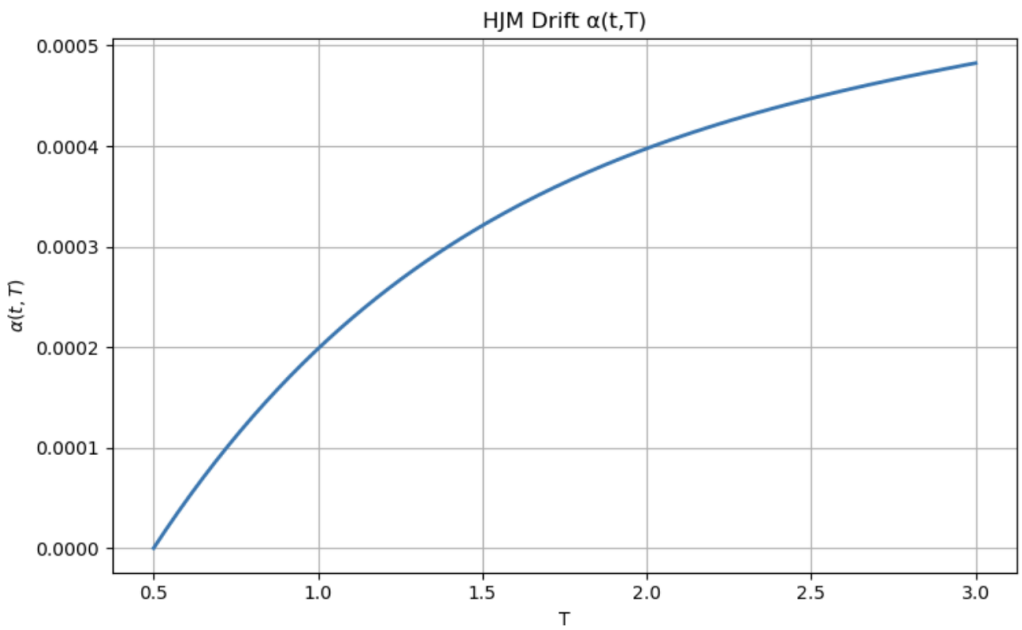

Figure 1: Forward-rate drift $\alpha(t,T)$ for $T \in [0.5, 3]$.

Effect of a on the volatility term structure. The bond-volatility vector is

$$v(t,T) = -\begin{pmatrix} v_1(t,T) \\ v_2(t,T) \end{pmatrix} = -\begin{pmatrix} \sigma_1(T-t) \\ \sigma_2\dfrac{1-e^{-a(T-t)}}{a} \end{pmatrix}.$$

The first component $v_1(t,T) = \sigma_1(T-t)$ is independent of a. The dependence on a comes entirely from

$$v_2(t,T) = \sigma_2\frac{1-e^{-a(T-t)}}{a}.$$

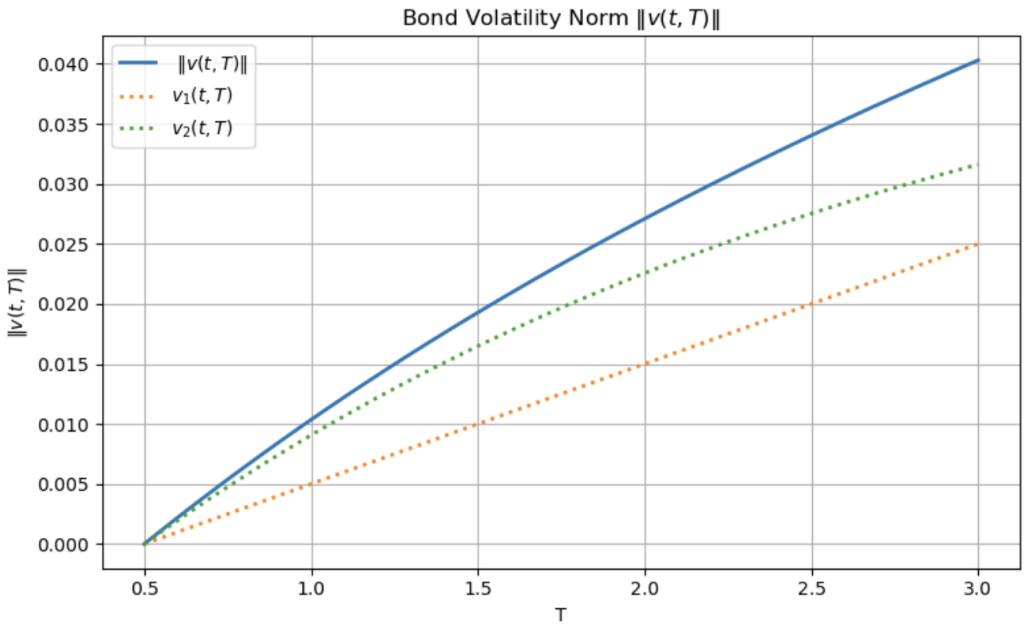

Figure 2: Bond volatility curves $\|v(t,T)\|$, $v_1(t,T)$ and $v_2(t,T)$ for $T \in [0.5, 3]$.

For $\tau = T - t > 0$,

$$\lim_{a\to 0} v_2(t,T) = \sigma_2 \tau, \qquad \lim_{a\to\infty} v_2(t,T) = 0.$$

Thus, for small a the second factor behaves like a non–mean-reverting volatility, affecting all maturities, while for large a its contribution is concentrated at very short maturities and becomes negligible for longer T. This is illustrated in Figure 3, where $v_2(t,T)$ is plotted for $a \in \{0.04, 0.4, 1, 4\}$.

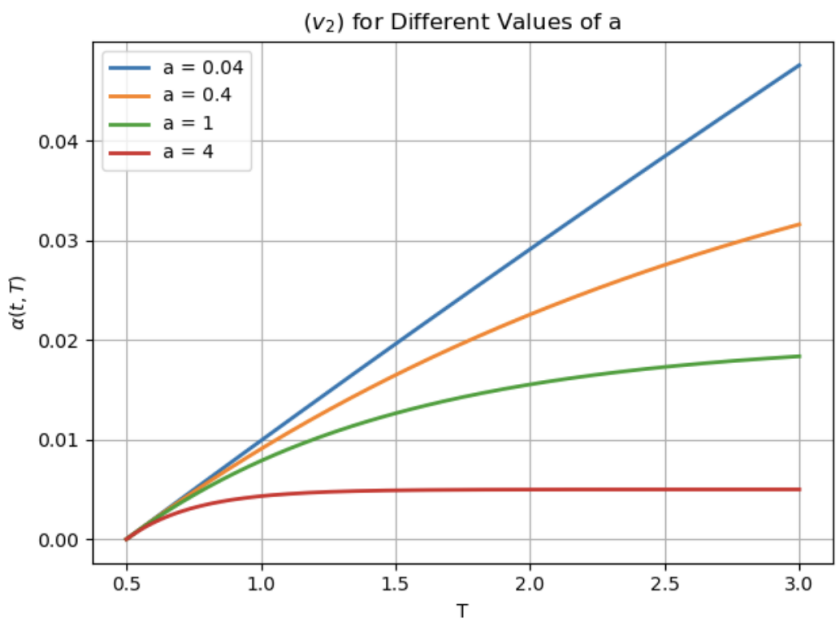

Figure 3: $v_2(t,T)$ for different values of a.

2 Exercise 2

2.a Question a

For $i = 1, 2$ we set

$$X_i(t) = \left(\sqrt{x_i} + B(t)\right)^2,$$

where $x = (x_1, x_2) \in \mathbb{R}_+^2$ is fixed and B is a one–dimensional Brownian motion.

Since $B(0) = 0$, we have

$$X_i(0) = \big(\sqrt{x_i} + B(0)\big)^2 = (\sqrt{x_i})^2 = x_i,$$

so

$$X(0) = (X_1(0), X_2(0)) = (x_1, x_2) = x.$$

SDE for each component. Fix $i \in \{1, 2\}$ and define

$$f(z) := \big(\sqrt{x_i} + z\big)^2, \qquad \text{so that } X_i(t) = f(B(t)).$$

By Itô's formula,

$$dX_i(t) = f'(B(t))\, dB(t) + \frac{1}{2} f''(B(t))\, dt,$$

and a quick computation gives

$$f'(z) = 2(\sqrt{x_i} + z), \qquad f''(z) = 2.$$

Hence

$$dX_i(t) = dt + 2\big(\sqrt{x_i} + B(t)\big)\, dB(t). \tag{1}$$

Note that

$$X_i(t) = \big(\sqrt{x_i} + B(t)\big)^2 \quad \Longrightarrow \quad \sqrt{X_i(t)} = \big|\sqrt{x_i} + B(t)\big|.$$

Introduce the process

$$W_i(t) := \int_0^t \operatorname{sgn}\big(\sqrt{x_i} + B(s)\big)\, dB(s),$$

where $\operatorname{sgn}(y) = 1$ if $y \geq 0$ and $\operatorname{sgn}(y) = -1$ if $y < 0$. It is standard that $(W_i(t))_{t \geq 0}$ is a Brownian motion, since it is a continuous martingale with quadratic variation

$$\langle W_i \rangle_t = \int_0^t \operatorname{sgn}\big(\sqrt{x_i} + B(s)\big)^2\, ds = \int_0^t 1\, ds = t.$$

In differential form, this means that

$$dW_i(t) = \operatorname{sgn}\big(\sqrt{x_i} + B(t)\big)\, dB(t).$$

and therefore,

$$\big(\sqrt{x_i} + B(t)\big)\, dB(t) = \big|\sqrt{x_i} + B(t)\big|\, dW_i(t) = \sqrt{X_i(t)}\, dW_i(t).$$

Substituting this into (1) yields

$$dX_i(t) = dt + 2\sqrt{X_i(t)}\, dW_i(t), \qquad X_i(0) = x_i, \quad i = 1, 2.$$

Instantaneous covariance matrix and non–affinity. From the original representation (1) with the common Brownian motion B, the diffusion vector is

$$\sigma(X(t), t) = 2 \begin{pmatrix} \sqrt{x_1} + B(t) \\ \sqrt{x_2} + B(t) \end{pmatrix}.$$

The instantaneous covariance matrix is

$$a(x, t) := \sigma(X(t), t)\, \sigma(X(t), t)^\top = 4 \begin{pmatrix} (\sqrt{x_1} + B(t))^2 & (\sqrt{x_1} + B(t))(\sqrt{x_2} + B(t)) \\ (\sqrt{x_1} + B(t))(\sqrt{x_2} + B(t)) & (\sqrt{x_2} + B(t))^2 \end{pmatrix}.$$

In particular, at time $t = 0$ we obtain

$$a(x, 0) = 4 \begin{pmatrix} x_1 & \sqrt{x_1 x_2} \\ \sqrt{x_1 x_2} & x_2 \end{pmatrix}.$$

For an affine diffusion on $\mathbb{R}_+^2$, each entry of the covariance matrix should be an affine function of the state $x = (x_1, x_2)$. Here the off–diagonal entry

$$a_{12}(x, 0) = 4\sqrt{x_1 x_2}$$

is clearly not affine in (x_1, x_2). Hence the diffusion matrix is not affine in x, and we conclude that X is **not an affine process**.

2.b Question b

For $u = (u_1, u_2) \in \mathbb{C}^2$ define

$$\varphi_t(u_1, u_2; x) := \mathbb{E}\big[e^{u_1 X_1(t) + u_2 X_2(t)}\big].$$

Recall

$$X_i(t) = (\sqrt{x_i} + B(t))^2, \qquad i = 1, 2,$$

and write $Y := B(t) \sim N(0, t)$. Then

$$X_i(t) = x_i + 2\sqrt{x_i}Y + Y^2,$$

so

$$u_1 X_1(t) + u_2 X_2(t) = aY^2 + 2bY + c,$$

where

$$a := u_1 + u_2, \qquad b := u_1\sqrt{x_1} + u_2\sqrt{x_2}, \qquad c := u_1 x_1 + u_2 x_2.$$

We complete the square:

$$aY^2 + 2bY = a\left(Y + \frac{b}{a}\right)^2 - \frac{b^2}{a},$$

so

$$u_1 X_1(t) + u_2 X_2(t) = a\left(Y + \frac{b}{a}\right)^2 + c - \frac{b^2}{a}.$$

Set

$$G := \frac{Y + b/a}{\sqrt{t}}.$$

Since $Y \sim N(0, t)$, we have $G \sim N\left(\frac{b}{a\sqrt{t}}, 1\right)$, hence $Z := G^2$ is noncentral χ^2 with $\delta = 1$ and noncentrality parameter

$$\zeta = \left(\frac{b}{a\sqrt{t}}\right)^2 = \frac{b^2}{a^2 t}.$$

Moreover

$$a\left(Y + \frac{b}{a}\right)^2 = atG^2 = atZ.$$

Therefore

$$\varphi_t(u_1, u_2; x) = e^{\,c - b^2/a}\,\mathbb{E}\big[e^{atZ}\big].$$

By the hint, for a noncentral χ^2-variable Z with δ degrees of freedom and noncentrality parameter ζ,

$$\mathbb{E}[e^{uZ}] = \frac{\exp\left(\dfrac{\zeta u}{1 - 2u}\right)}{(1 - 2u)^{\delta/2}}, \qquad u \in \mathbb{C}_-.$$

Here $\delta = 1$, $\zeta = b^2/(a^2 t)$ and $u = at$, so

$$\mathbb{E}\big[e^{atZ}\big] = \frac{\exp\left(\dfrac{b^2/a}{1 - 2at}\right)}{\sqrt{1 - 2at}},$$

because $\zeta at = \frac{b^2}{a^2 t} at = b^2/a$.

Hence

$$\varphi_t(u_1, u_2; x) = \frac{1}{\sqrt{1 - 2at}}\,\exp\left(c - \frac{b^2}{a} + \frac{b^2/a}{1 - 2at}\right).$$

A short simplification gives

$$c - \frac{b^2}{a} + \frac{b^2/a}{1 - 2at} = c + \frac{2tb^2}{1 - 2at},$$

so we can write the characteristic function as

$$\boxed{\varphi_t(u_1,u_2;x) = \frac{1}{\sqrt{1-2(u_1+u_2)t}}\,\exp\!\left(u_1x_1+u_2x_2+\frac{2t\,(u_1\sqrt{x_1}+u_2\sqrt{x_2})^2}{1-2(u_1+u_2)t}\right),}$$

for those u such that $\mathrm{Re}(1-2(u_1+u_2)t)>0$.

Taking the logarithm,

$$\log\varphi_t(u_1,u_2;x) = -\tfrac{1}{2}\log\big(1-2(u_1+u_2)t\big)+u_1x_1+u_2x_2+\frac{2t\,(u_1\sqrt{x_1}+u_2\sqrt{x_2})^2}{1-2(u_1+u_2)t}.$$

The last term contains

$$\frac{4t\,u_1u_2}{1-2(u_1+u_2)t}\sqrt{x_1x_2},$$

which is not of the form $\psi_1(t,u)x_1+\psi_2(t,u)x_2$. So $\log\varphi_t(u_1,u_2;x)$ is not affine in x, and the characteristic function is not exponential–affine. This agrees with part (a): X is not an affine process.

2.c Question c

We consider

$$dY_1(t) = \big(b_1+\beta_{11}Y_1(t)+\beta_{12}Y_2(t)\big)\,dt+\sigma_1\sqrt{Y_1(t)}\,dW_1(t),$$
$$dY_2(t) = \big(b_2+\beta_{21}Y_1(t)+\beta_{22}Y_2(t)\big)\,dt+\sigma_2\sqrt{Y_2(t)}\,dW_2(t),$$

where W_1 and W_2 are independent Brownian motions.

The diffusion matrix of $Y=(Y_1,Y_2)^\top$ is

$$\sigma(Y) = \begin{pmatrix}\sigma_1\sqrt{Y_1} & 0\\ 0 & \sigma_2\sqrt{Y_2}\end{pmatrix},$$

so

$$a(Y) := \sigma(Y)\sigma(Y)^\top = \begin{pmatrix}\sigma_1^2Y_1 & 0\\ 0 & \sigma_2^2Y_2\end{pmatrix}.$$

For X we found in part (a)

$$a(X) = 4\begin{pmatrix}X_1 & \sqrt{X_1X_2}\\ \sqrt{X_1X_2} & X_2\end{pmatrix},$$

which contains the non–affine term $4\sqrt{X_1X_2}$ in the off–diagonal entries.

For Y, the drift is

$$b(Y) = \begin{pmatrix}b_1\\ b_2\end{pmatrix}+\begin{pmatrix}\beta_{11} & \beta_{12}\\ \beta_{21} & \beta_{22}\end{pmatrix}\begin{pmatrix}Y_1\\ Y_2\end{pmatrix},$$

which is affine in Y, and each entry of $a(Y)$ is of the form $\alpha_{ij}+\sum_k\beta_{ij,k}Y_k$ (here $\alpha_{ij}=0$). Hence Y is an affine diffusion, in contrast to X.

From a modelling point of view, working with Y preserves the exponential–affine form of the characteristic function and the usual tractability of affine models (riccati equations, Fourier pricing, etc.), while using X would destroy this tractability and typically require heavier numerical methods.

3 Exercise 3

(a) Discounted price is a non–negative martingale

Consider the affine stock price model with dynamics:

$$dS(t) = rS(t)dt+\sigma\sqrt{S(t)}dW(t),\quad S(0)=s_0\geq 0.$$

Let $f(t,S) = e^{-rt}S$. By Itô's lemma,

$$\begin{aligned} d(e^{-rt}S_t) &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S_t}dS_t + \frac{1}{2}\frac{\partial^2 f}{\partial S_t^2}\sigma^2 S_t dt \\ &= -re^{-rt}S_t dt + e^{-rt}dS_t \\ &= -re^{-rt}S_t dt + e^{-rt}(rS_t dt + \sigma\sqrt{S_t}dW_t) \\ &= e^{-rt}\sigma\sqrt{S_t}dW_t \end{aligned}$$

Hence $\tilde{S}_t := e^{-rt}S_t$ is a continuous local martingale. To upgrade it to a true martingale, check square–integrability of the integrand:

$$\mathbb{E}\Big[\int_0^t (e^{-rs}\sigma\sqrt{S_s})^2 ds\Big] = \sigma^2\int_0^t e^{-2rs}\,\mathbb{E}[S_s]\,ds = \sigma^2 S_0\int_0^t e^{-rs}\,ds = \begin{cases} \dfrac{\sigma^2 S_0}{r}\big(1-e^{-rt}\big), & r>0, \\ \sigma^2 S_0\, t, & r=0, \end{cases}$$

which is finite. Therefore $\tilde{S}_t$ is a martingale and $\mathbb{E}[\tilde{S}_t \mid \mathcal{F}_s] = \tilde{S}_s$, $\mathbb{E}[\tilde{S}_t] = S_0$. Since the diffusion coefficient $\sigma\sqrt{S_t}$ vanishes at 0, S_t stays ≥ 0 and 0 is absorbing. Thus $e^{-rt}S_t$ is a non–negative martingale.

(b) Exponential–affine martingale and the law of Z_t

Fix $u \in \mathbb{C}$ with $\mathrm{Re}(u) < \frac{1}{2}$ and define

$$C(t) = \frac{\sigma^2}{4r}\big(e^{rt}-1\big), \qquad M_s := \exp\big(\alpha(s) + \beta(s)S_s\big), \quad s \in [0,t],$$

with terminal conditions $\alpha(t) = 0$, $\beta(t) = u/C(t)$.

$$\begin{aligned} \text{Applying Itô to } M_s: \quad dM_s &= \frac{\partial M_s}{\partial s}ds + \frac{\partial M_s}{\partial S_s}dS_s + \frac{1}{2}\frac{\partial^2 M_s}{\partial S_s^2}\sigma^2 S_s dt \\ \text{divide by } M_s: \quad \frac{dM_s}{M_s} &= (\alpha'(s) + \beta'(s)S_s)ds + \beta(s)(rS_s ds + \sigma\sqrt{S_s}dW_s) + \frac{1}{2}\beta^2(s)\sigma^2 S_s ds \\ &= (\alpha'(s) + \beta'(s)S_s + r\beta(s)S_s + \frac{1}{2}\beta^2(s)\sigma^2 S_s)ds + \beta(s)\sigma\sqrt{S_s}dW_s \end{aligned}$$

We choose α, β to kill the drift for all S_s:

$$\alpha'(s) = 0, \qquad \beta'(s) + r\beta(s) + \tfrac{1}{2}\sigma^2\beta(s)^2 = 0,$$

so $\alpha \equiv 0$ and β solves a Riccati ODE. Set $g(s) := 1/\beta(s) \Rightarrow g' = -\frac{\beta'(s)}{\beta^2(s)}$, then

$$g'(s) - rg(s) = \tfrac{1}{2}\sigma^2, \qquad g(t) = \frac{C(t)}{u}.$$

This is a linear ODE. Multiply by integrating factor e^{-rs}:

$$\frac{d}{ds}\left(e^{-rs}g\right) = -re^{-rs} + e^{-rs}g' = \frac{1}{2}\sigma^2 e^{-rs}, \quad \Rightarrow \int_s^t \frac{d}{du}\left(e^{-ru}g(u)\right)du = \int_s^t \frac{1}{2}\sigma^2 e^{-ru}du,$$

$$\Rightarrow e^{-rt}g(t) - e^{-rs}g(s) = \frac{1}{2r}\sigma^2(e^{-rs} - e^{-rt}).$$

Using $g(t) = \frac{C(t)}{u}$, we get:

$$g(s) = \frac{1}{2r}\sigma^2\left(e^{r(s-t)} - 1\right) + e^{r(s-t)}\frac{C(t)}{u}. \quad \Rightarrow \quad \beta(s) = \frac{1}{g(s)} = \left(\frac{1}{2r}\sigma^2\left(e^{r(s-t)} - 1\right) + e^{r(s-t)}\frac{C(t)}{u}\right)^{-1}.$$

This simplifies to :

$$\beta(s) = \frac{e^{r(t-s)}}{1 - 2\frac{C(t-s)}{C(t)}u}\frac{u}{C(t)}$$

In particular, $\beta(0) = \dfrac{(u/C(t))\, e^{rt}}{1-2u}$.

Because M_s is a martingale, $\mathbb{E}[M_t] = \mathbb{E}[M_0]$, i.e.

$$\mathbb{E}\Big[e^{\frac{u}{C(t)}S_t}\Big] = \exp\big(\beta(0)S_0\big) = \exp\left(\frac{e^{rt}S_0}{C(t)}\frac{u}{1-2u}\right).$$

Thus $Z(t) := S(t)/C(t)$ has the noncentral χ^2 mgf:

$$M(u;k,\lambda) = \frac{\exp\left[\frac{\lambda u}{1-2u}\right]}{(1-2u)^{k/2}}$$

with $k = 0$ df and noncentrality $\lambda = \dfrac{e^{rt}S_0}{C(t)}$.

(c) Financial interpretation of the process

Since the diffusion coefficient $\sigma\sqrt{S_t}$ vanishes at $S_t = 0$, the process S_t can reach zero in finite time and, once it does, it remains there forever — i.e. the point 0 is an **absorbing state**. In financial terms, this means that the asset price process represents a **defaultable equity with zero recovery**: once the firm defaults, the stock price collapses to 0 and stays there.

This feature makes the model suitable for valuing instruments whose payoff depends on the possibility of such absorption, for instance:

- equity or firm value processes in **structural credit-risk models**, where hitting zero corresponds to bankruptcy,
- **equity-linked notes** or options that knock out upon default,
- or any contingent claim with **zero recovery at default**.

Unlike the geometric Brownian motion, which never reaches zero in finite time, the square–root diffusion captures the risk of default endogenously. Thus, the process provides a parsimonious model for equities or assets subject to default risk, where the hitting time of zero plays the role of the default time $\tau = \inf\{t \geq 0 : S_t = 0\}$.

(d) Call price in terms of noncentral χ^2 tails

With $c := C(T)$ and $Z := Z(T) = S(T)/c$, $Z \sim \chi_0'^2(\lambda)$, $\lambda = \dfrac{e^{rT}S_0}{c}$. Then

$$\text{Call}(0) = e^{-rT}\mathbb{E}\big[(S(T)-K)^+\big] = e^{-rT}\big(c\,\mathbb{E}[(Z-k)^+]\big), \quad k := K/c.$$

Using

$$\mathbb{P}(Z > k) = \bar{F}_{0,\lambda}(k) \quad \text{and} \quad \mathbb{E}[Z\,\mathbf{1}_{\{Z>k\}}] = \lambda\,\bar{F}_{2,\lambda}(k)$$

,

$$\begin{aligned}\text{Call}(0) &= e^{-rT}\big(c\lambda\bar{F}_{2,\lambda}(k) - K\bar{F}_{0,\lambda}(k)\big)\\ &= e^{-rT}\big(e^{rT}S_0\bar{F}_{2,\lambda}(k) - K\bar{F}_{0,\lambda}(k)\big)\\ &= S_0\,\bar{F}_{2,\lambda}\left(\frac{K}{c}\right) - e^{-rT}K\,\bar{F}_{0,\lambda}\left(\frac{K}{c}\right).\end{aligned}$$

where $\bar{F}_{\mu,\lambda} = 1 - F_{\mu,\lambda}$ is the survival function with parameters μ and λ

Computing this using Python for: $r = 5\%, \quad \sigma = 40\%, \quad s_0 = 100, \quad T = 1, \quad K \in \{60, 80, 100, 120, 140\}$, we get the following results:

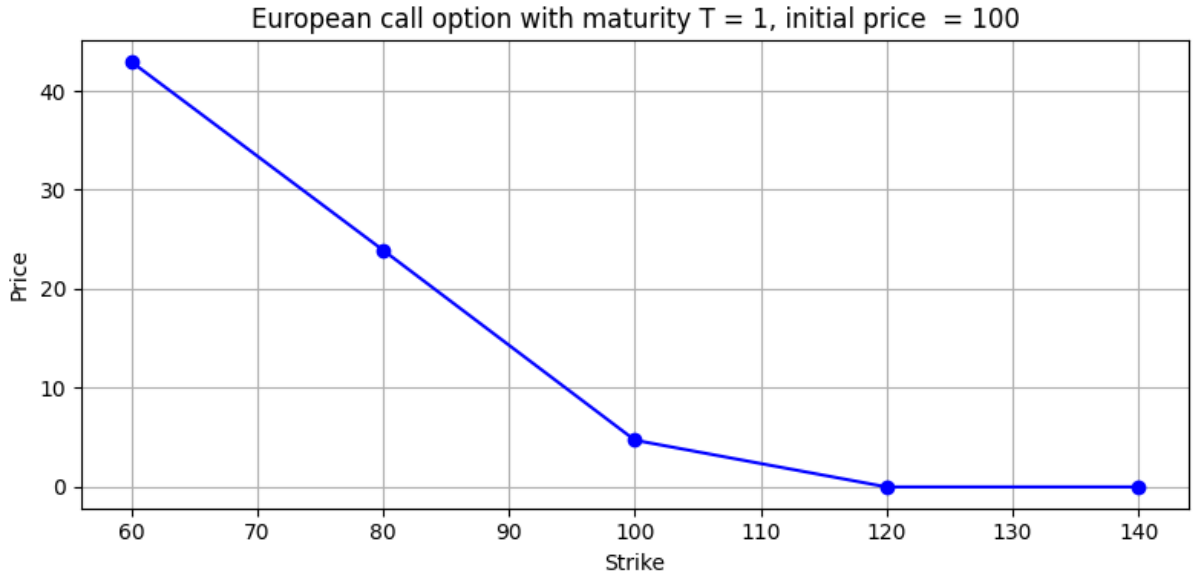

Figure 4: European call option with maturity $T = 1$, initial price $S_0 = 100$.

Discussion. The call price grows monotonically with the initial price S_0, as a larger S_0 directly raises the expected level of S_T and the noncentrality parameter λ, increasing the probability that the option finishes in the money.

The dependence on the ratio $\frac{r}{\sigma}$ is more complex because the two parameters have conflicting effects:

- **Interest rate (r):** An increase in r raises the ratio $\frac{r}{\sigma}$ and the drift. This increases the value of the call option (Rho > 0).
- **Volatility (σ):** A decrease in σ also raises the ratio $\frac{r}{\sigma}$ and increases λ (implying a lower probability of absorption at zero). However, a lower σ reduces the variance of the terminal distribution. Due to the convex payoff of a call option, the reduction in upside potential typically outweighs the benefit of increased survival probability, leading to a lower option price (Vega > 0).

In summary, while the survival probability increases with $\frac{r}{\sigma}$, the option price generally benefits from higher variance:

$$\boxed{\text{Higher } S_0, \text{ higher } r, \text{ or higher } \sigma \text{ (lower } \tfrac{r}{\sigma}) \;\Rightarrow\; \text{higher call price.}}$$

4 Exercise 4

As in question 1, we work under the two–factor Gaussian HJM model. It is characterized by :

- a flat initial curve $f(0, T) = 5\%$
- $(a, \sigma_1, \sigma_2) = (0.4, 0.005, 0.008)$

The forward dynamics and drift are given as before by :

$$df(t, T) = \alpha(t, T)\, dt + \sigma_1\, dW_1^*(t) + \sigma_2 e^{-a(T-t)}\, dW_2^*(t)$$

$$\alpha(t, T) = \sigma_1^2 (T - t) + \sigma_2^2 e^{-a(T-t)} \frac{1 - e^{-a(T-t)}}{a}.$$

4.a Simulation of $f(t,T)$ and bond prices

To compute the forward-rate and the tero coupon bond price, we use a time grid $t_n = n\Delta t$, $\Delta t = 1/252$, $n = 0, \dots, N_{\text{steps}}$ up to a year since we simulate on $t \in \{0, 1\}$.

To simulate the data, we use a Euler scheme given by :

$$\begin{cases} f_{n+1}(T_j) = f_n(T_j) + \alpha(t_n, T_j)\,\Delta t + \sigma_1\,\Delta W_{1,n} + \sigma_2 e^{-a(T_j - t_n)}\Delta W_{2,n}, \\ \quad f_0(T_j) = 0.05. \end{cases}$$

The two brownian motions are simulated using :

$$\Delta W_{k,n} = \sqrt{\Delta t} Z_{k,n}, \qquad Z_{k,n} \sim \mathcal{N}(0,1), \qquad k = 1, 2$$

where we used $N = 10,000$ paths.

By definition of instantaneous forward rates we know that they are connected to bond prices via the following equation :

$$P(t_n, T_j) = \exp\Big(- \int_{t_n}^{T_j} f(t_n, u)\, du \Big)$$

If we wanted to compute this integral, we would've need the entire forward curve $u \to f(t_n, u)$ for $u \in [t, T_j]$. Unfortunately, we are limited in our simulation by the number of maturities we have. Therefore we will use the approximation by trapezoids on the maturity grid $\{T_k\}$:

Let m_0 be the first index such that $T_{m_0} \geq t_n$.

$$\begin{cases} \displaystyle\int_{t_n}^{T_j} f(t_n, u)\, du \;\approx\; (T_{m_0} - t_n)\, f(t_n, T_{m_0}) + \sum_{k=m_0+1}^{j} \tfrac{1}{2}\,(T_k - T_{k-1})\big(f(t_n, T_{k-1}) + f(t_n, T_k)\big), \\ \displaystyle P(t_n, T_j) \;\approx\; \exp\Big(-\,(T_{m_0} - t_n)\, f(t_n, T_{m_0}) - \sum_{k=m_0+1}^{j} \tfrac{1}{2}\,(T_k - T_{k-1})\big(f(t_n, T_{k-1}) + f(t_n, T_k)\big)\Big), \\ \displaystyle P(0, T_j) \;=\; \exp\Big(-\int_0^{T_j} f_0(u)\, du \Big) \;=\; \exp(-0.05\, T_j). \end{cases}$$

which is consistent with the Euler update.

Plots
For each $T_j = \{0.5, 1, 2\}$, The plots are shown below.

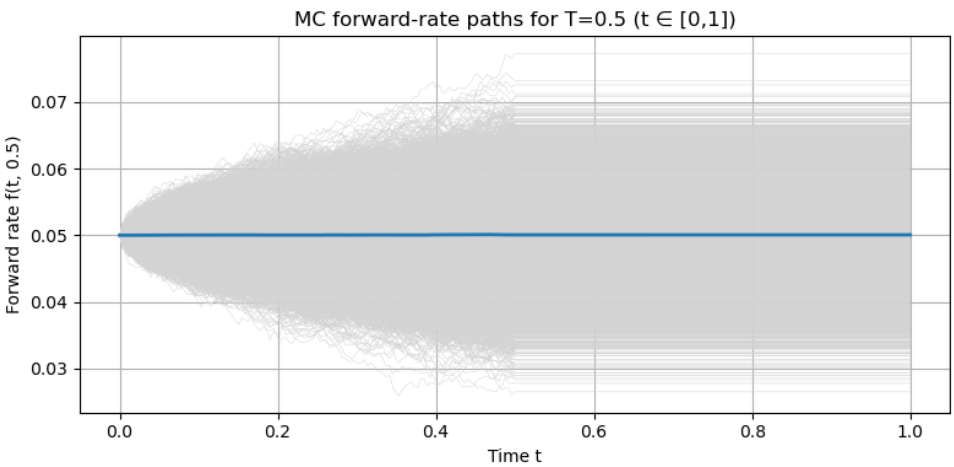

Figure 5: Monte Carlo forward-rate paths $f(t, 0.5)$

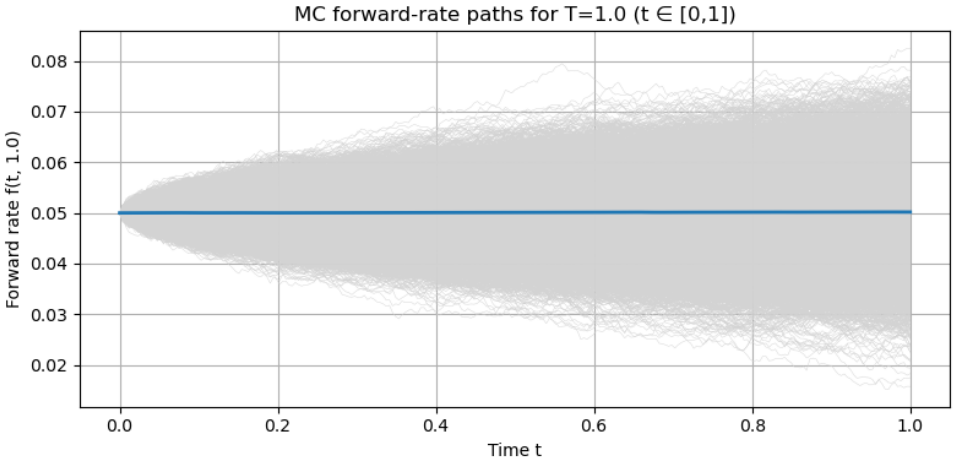

Figure 6: Monte Carlo forward-rate paths $f(t, 1)$

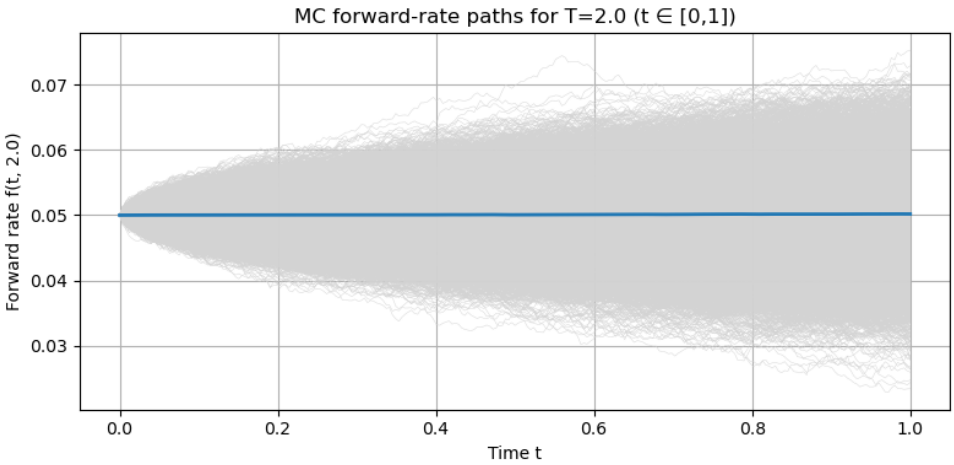

Figure 7: Monte Carlo forward-rate paths $f(t, 2)$

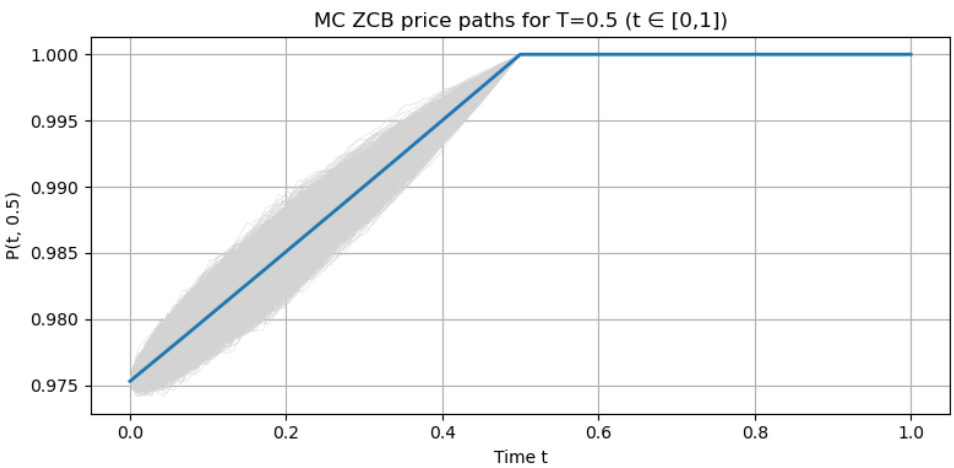

Figure 8: Monte Carlo zero coupon bond $P(t, 0.5)$

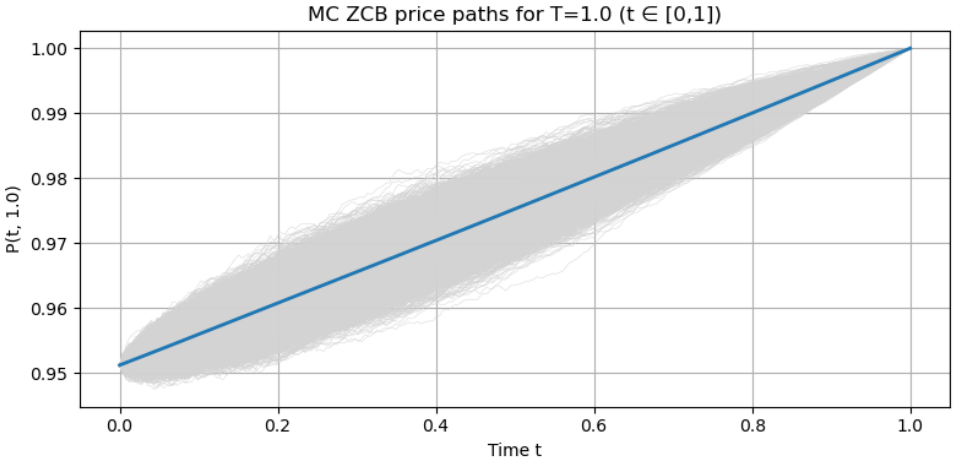

Figure 9: Monte Carlo zero coupon bond $P(t, 1)$

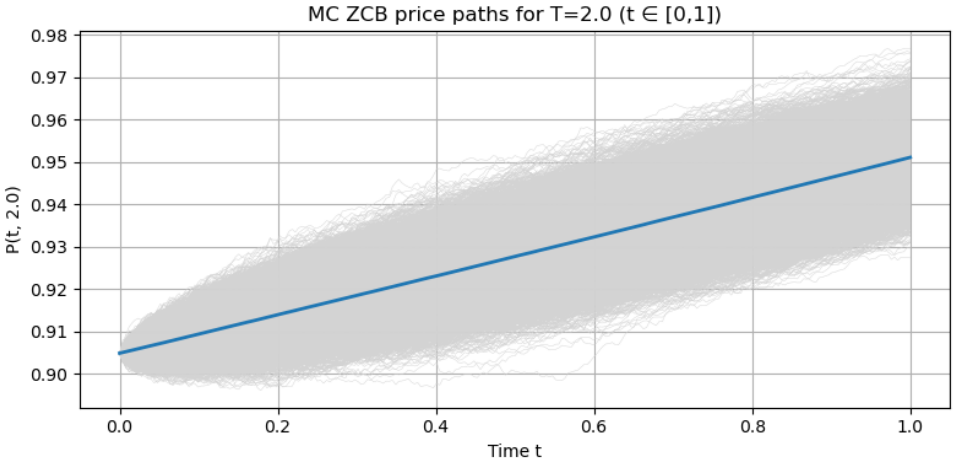

Figure 10: Monte Carlo zero coupon bond $P(t, 2)$

Each light–grey curve represents one simulated Monte Carlo path; the blue curve is the empirical mean across all simulated paths at each time t.

Forward rate $f(t, T_j)$
For $T = 0.5$ the forward rate stops evolving once t passes 0.5 because the time–to–maturity $\tau = T - t$ becomes non–positive, so the Euler update freezes and the paths remain flat for $t \geq 0.5$.
For $T = 1$ and $T = 2$ the dispersion widens over $t \in [0, 1]$ as randomness accumulates while $\tau > 0$.

Zero Coupon Bond $P(t, T_j)$
As time advances, $P(t, T)$ increases because the remaining discounting horizon $T - t$ shrinks. For $T = 0.5$, all paths hit $P(t, 0.5) = 1$ exactly at $t = 0.5$ and then remain flat for $t \geq 0.5$ (the bond has matured).
For $T = 1$ the price rises from $P(0, 1) \approx e^{-0.05} \approx 0.95123$ to 1 at $t = 1$.
For $T = 2$ the price rises from $P(0, 2) \approx e^{-0.05\times 2} \approx 0.90484$ to 1 at $t = 1$. Therefore since we only simulate until $t = 1$ we can't see the full convergence on the plot.
The spread of the grey fan reflects path variability; it is larger for longer maturities since there is more time for shocks to accumulate before reaching T.

4.b Forward and futures rate

The simple forward rate at $t = 0$ is given by:

$$F(0, 0.5, 1) = \frac{1}{0.5}\left(\frac{P(0, 0.5)}{P(0, 1)} - 1\right)$$

Given the expression of $P(0,T)$ earlier, we have :

$$F(0,0.5,1) = \frac{1}{0.5}\left(\frac{P(0,0.5)}{P(0,1)} - 1\right) = \frac{1}{0.5}\left(\frac{e^{-0.05\times 0.5}}{e^{-0.05\times 1}} - 1\right) = 0.05063$$

Therefore the value of the simple forward rate at t=0 is approximately **5.063%**

The futures rate is given by :

$$\mathbb{E}^{\mathbb{Q}}\big[LR(0.5,1)\big] = \mathbb{E}^{\mathbb{Q}}\left(\frac{1}{0.5}\left(\frac{1}{P(0.5,1)} - 1\right)\right)$$

With N Monte Carlo paths and $P^{(i)}(0.5,1)$ from path i,

$$\widehat{\mathbb{E}}_N^{\mathbb{Q}}\big[LR(0.5,1)\big] = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{0.5}\left(\frac{1}{P^{(i)}(0.5,1)} - 1\right).$$

Where each $P^{(i)}(0.5,1)$ is defined above.

Therefore we have : $\mathbb{E}^{\mathbb{Q}}\big[LR(0.5,1)\big] = 0.05066$ and the future rate is **5.066%**

Discussion

In our simulation, the futures rate slightly exceeds the forward (0.05066 vs. 0.05063), a difference of $0.00003 \approx 0.3$ bp. This gap is the classic convexity adjustment that arises when interest rates (and hence bond prices) are stochastic. The futures rate is the risk-neutral **undiscounted** expectation of the settlement rate due to daily mark-to-market, whereas the forward embeds discounting under random rates. Since

$$LR(0.5,1) = \frac{1}{0.5}\left(\frac{1}{P(0.5,1)} - 1\right)$$

is convex in $1/P(0.5,1)$, Jensen's inequality implies $\mathbb{E}[LR(0.5,1)]$ typically exceeds the forward rate when there is volatility in $P(0.5,1)$. The adjustment is small here because the horizon is short and volatilities are moderate.

From the change-of-numéraire argument used in the lectures, one can write

$$1 + \delta\, LR_{\text{fut}}(t;T_0,T_1) = \big(1 + \delta\, F(t;T_0,T_1)\big) \times (\text{adjustment factor}),$$

where the adjustment factor is a conditional expectation under the T_1-forward measure that depends on the HJM volatility kernel. A key implication is that if

$$\sigma(s,v)\,\sigma(s,u)^{\top} \;\geq 0 \quad \text{for all } s,u,v,$$

then the adjustment factor is ≥ 1 and hence $LR_{\text{fut}}(t;T_0,T_1) \geq F(t;T_0,T_1)$. In our two-factor Gaussian HJM, $\Sigma(s,u) = (\sigma_1,\ \sigma_2 e^{-a(u-s)})^{\top}$ is deterministic and yields a non-negative kernel, so the condition is satisfied. This explains why we observe a (small) positive convexity premium in the numbers above.

4.c Caplet pricing for fixed strike ($K = 5\%$)

We price quarterly caplets at maturities $T_i = i/4$ for $i = 1,\ldots,8$, with accrual $\delta_i = T_i - T_{i-1} = 0.25$. The caplet payoff at T_i is

$$\text{Cpl}_i(K) = \delta_i\ \max\big(LR(T_{i-1},T_i) - K, 0\big) = \max\left(\frac{1}{P(T_{i-1},T_i)} - (1 + K\delta_i),\ 0\right).$$

Under the T_i-forward measure $\mathbb{Q}^{T_i}$, $LR(\cdot;T_{i-1},T_i)$ is a martingale and discounting is by $P(0,T_i)$.

The caplet payoff at T_i is

$$\text{Cpl}_i(K) = \delta_i\ \max\big(LR(T_{i-1},T_i) - K, 0\big) = \max\left(\frac{1}{P(T_{i-1},T_i)} - (1 + K\delta_i),\ 0\right).$$

Under the T_i–forward measure $\mathbb{Q}^{T_i}$, the numéraire is $P(\cdot,T_i)$, so $L(\cdot;T_{i-1},T_i)$ is a martingale and discounting is by $P(0,T_i)$. It is convenient to define

$$X := \frac{1}{P(T_{i-1},T_i)}, \qquad K_X := 1 + K\delta_i,$$

so that $\mathrm{Cpl}_i(K) = \max(X - K_X, 0)$ and

$$\mathrm{Price}_0\big(\mathrm{Cpl}_i(K)\big) = P(0,T_i)\,\mathbb{E}^{\mathbb{Q}^{T_i}}\left[\max(X - K_X, 0)\right].$$

Gaussian HJM dynamics and lognormality of X.

We work in the two–factor Gaussian HJM used throughout the report. The instantaneous forward–rate volatility is :

$$\Sigma(t,T) = \begin{pmatrix} \sigma_1 \\ \sigma_2\, e^{-a\,(T-t)} \end{pmatrix},$$

Bond prices under $\mathbb{Q}^{T_i}$ satisfy :

$$\frac{dP(t,T)}{P(t,T)} \;=\; -\left(\int_t^T \Sigma(t,u)\,du\right)^{\top} dW_t^{T_i}, \qquad 0 \le t \le T \le T_i,$$

where W^{T_i} is a 2–dimensional Brownian motion under $\mathbb{Q}^{T_i}$. Therefore the **bond ratio** $P(t,T_{i-1})/P(t,T_i)$ is a $\mathbb{Q}^{T_i}$–martingale driven by:

$$d\log\Big(\tfrac{P(t,T_{i-1})}{P(t,T_i)}\Big) = -\left(\int_t^{T_{i-1}} \Sigma(t,u)\,du - \int_t^{T_i} \Sigma(t,u)\,du\right)^{\top} dW_t^{T_i}.$$

Integrating from 0 to T_{i-1} shows that $X = P(T_{i-1},T_{i-1})/P(T_{i-1},T_i) = 1/P(T_{i-1},T_i)$ is lognormal under $\mathbb{Q}^{T_i}$ with

$$\mathbb{E}^{\mathbb{Q}^{T_i}}[X] = X_0 := \frac{P(0,T_{i-1})}{P(0,T_i)}, \qquad \log X \sim \mathcal{N}\big(\log X_0 - \tfrac{1}{2}V_i,\; V_i\big),$$

where the time–0 log–variance is

$$V_i = \int_0^{T_{i-1}} \left\| \int_{T_{i-1}}^{T_i} \Sigma(s,u)\,du \right\|^2 ds.$$

With our $\Sigma(s,u) = (\sigma_1,\; \sigma_2 e^{-a(u-s)})^{\top}$, the inner integral is

$$\int_{T_{i-1}}^{T_i} \Sigma(s,u)\,du = \begin{pmatrix} \sigma_1\,\delta_i \\ \sigma_2 \displaystyle\int_{T_{i-1}}^{T_i} e^{-a(u-s)}\,du \end{pmatrix} = \begin{pmatrix} \sigma_1\,\delta_i \\ \dfrac{\sigma_2}{a}\big(1 - e^{-a\delta_i}\big)\,e^{-a(T_{i-1}-s)} \end{pmatrix}.$$

Squaring and integrating over $s \in [0, T_{i-1}]$ yields

$$\boxed{V_i = \sigma_1^2\,\delta_i^2\,T_{i-1} \;+\; \frac{\sigma_2^2\big(1 - e^{-a\delta_i}\big)^2}{a^2} \cdot \frac{1 - e^{-2aT_{i-1}}}{2a}.}$$

This is the variance used in our code.

Black price

Since X is lognormal under $\mathbb{Q}^{T_i}$ with forward X_0 and total variance V_i, the caplet admits the Black representation

$$C_i^{\mathrm{Blk}}(K) = P(0,T_i)\,\big[X_0\,\Phi(d_1) - K_X\,\Phi(d_2)\big], \qquad d_{1,2} = \frac{\ln(X_0/K_X) \pm \frac{1}{2}V_i}{\sqrt{V_i}}.$$

Results

Table 2: Caplet prices by maturity ($K = 5\%$, $(a, \sigma_2) = (0.4, 0.008)$)

T_i	Caplet_Black
0.25	0.00008
0.50	0.00047
0.75	0.00063
1.00	0.00073
1.25	0.00080
1.50	0.00086
1.75	0.00091
2.00	0.00094

4.d ATM caplets and Black implied volatilities

ATM caplets pricing

For each i, we can define the ATM strike by :

$$K_i^{\text{ATM}} = \frac{1}{\delta_i}\left(\frac{P(0,T_{i-1})}{P(0,T_i)} - 1\right) = \frac{1}{\delta_i}\left(e^{-0.05(T_{i-1}-T_i)} - 1\right) \approx 0.05031$$

ATM caplets are priced by the same method (c), but here we replace $K = K_i^{\text{ATM}}$:

The caplets prices with a At The Money strike are therefore given by :

Table 3: ATM caplet prices $\widehat{C}_i(K_i^{\text{ATM}})$

T_i	$\widehat{C}_i(K_i^{\text{ATM}})$
0.25	0.00000
0.50	0.00043
0.75	0.00059
1.00	0.00069
1.25	0.00076
1.50	0.00082
1.75	0.00087
2.00	0.00091

Edge case $i = 1$.

For the first bucket $T_0 = 0$, $V_1 = 0$ and the caplet is deterministic:

$$C_1^{\text{Blk}}(K) = P(0,T_1)\,\max\Big(\tfrac{P(0,0)}{P(0,T_1)} - (1 + K\delta_1),\, 0\Big),$$

Black Scholes implied volatility

The Caplet prices is also given by Black Scholes formula by :

$$C_i^{\text{BS}}(\sigma) = P(0,T_i)\,\delta_i\big(F_i N(d_1) - K N(d_2)\big), \quad F_i = \frac{1}{\delta_i}\left(\frac{P(0,T_{i-1})}{P(0,T_i)} - 1\right),$$

with :

$$d_{1,2} = \frac{\ln(F_i/K) \pm \frac{1}{2}\sigma^2\tau_i}{\sigma\sqrt{\tau_i}}$$

and $\tau_i = T_{i-1}$.

We seek the volatility σ that makes the Black caplet price match the caplet prices that we computed before price.

$$C_i^{BS}(\sigma) = \widehat{C}_i(K_i), \qquad C_i^{BS}(\sigma_{ATM}) = \widehat{C}_i(K_i^{\text{ATM}})$$

To do so,

- **Limit the target price** to a valid range: between intrinsic $P(0,T_i)\delta \max(F-K,0)$ (a caplet can never be worth less than it's discounted value to avoid arbitrage) and a safe upper bound $P(0,T_i)\delta F$ (the present value of the maximum payoff possible).
- **Root-finding :** define $f(\sigma) = C^{\text{Black}}(\sigma) - \text{Price}_{\text{target}}$ and find $\sigma > 0$ such that $f(\sigma) = 0$.
- **Robust solve:** bracket a sign change for f and use a bracketing solver (Brent). If needed, fall back to a safeguarded Newton step using vega.

Key idea: implied volatility is the σ that reconciles the Black model with the Monte Carlo price. No model re-fitting, just solve price equality for σ with bounds that ensure a feasible solution.

Table 4: Black implied volatilities : fixed strike $K = 5\%$ vs. ATM, rounded to 5 decimals

T_i	$\sigma_{\text{imp}}^{\text{Black}}(K{=}5\%)$	$\sigma_{\text{imp}}^{\text{Black}}(\text{ATM})$
0.25	0.00000	0.00000
0.50	0.17781	0.17726
0.75	0.17232	0.17178
1.00	0.16733	0.16681
1.25	0.16280	0.16230
1.50	0.15870	0.15821
1.75	0.15497	0.15449
2.00	0.15158	0.15111

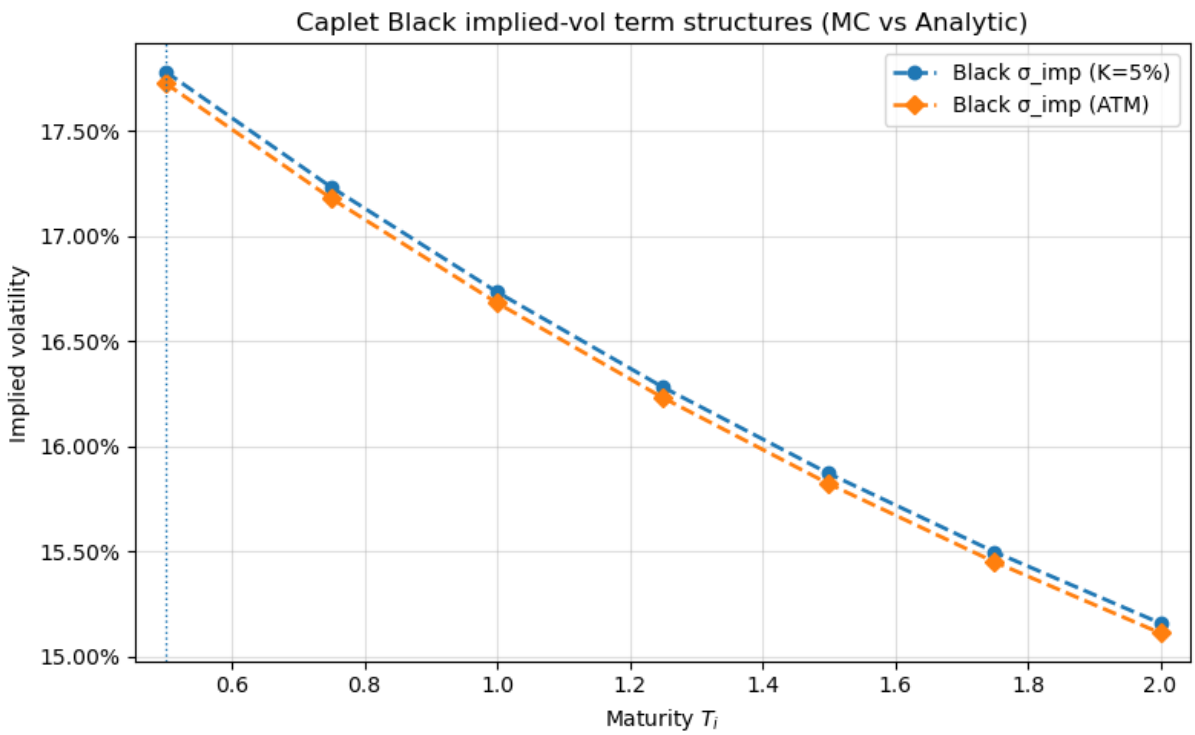

Figure 11: Caplet Black implied volatility term structures

4.e Parameter study

We use the same workflow as in (b)–(d), only parameters are changed and the random seeds/path counts kept comparable.

Note that the simple forward rate and the ATM strike is kept the same even with these parameter change since they are only influenced by the delta of maturity and the flat initial curve.

4.e.1 $(a, \sigma_2) = (0.2, 0.012)$

The future rate is given by : $\mathbb{E}^{\mathbb{Q}}\big[LR(0.5, 1)\big] = 0.05066$

The Caplets prices for each strike and the corresponding implied volatilities are given by

Table 5: Caplet prices and Black implied volatilities ($(a, \sigma_2) = (0.2, 0.012)$)

T_i	$\widehat{C}_i(K{=}5\%)$	$\widehat{C}_i(K_{\text{ATM}})$	$\sigma_{\text{imp}}(K{=}5\%)$	$\sigma_{\text{imp}}(K_{\text{ATM}})$
0.25	0.00008	0.00000	0.00000	0.00000
0.50	0.00065	0.00061	0.25181	0.25103
0.75	0.00088	0.00084	0.24688	0.24611
1.00	0.00103	0.00100	0.24216	0.24141
1.25	0.00115	0.00111	0.23764	0.23690
1.50	0.00124	0.00121	0.23331	0.23259
1.75	0.00132	0.00128	0.22917	0.22846
2.00	0.00138	0.00134	0.22520	0.22450

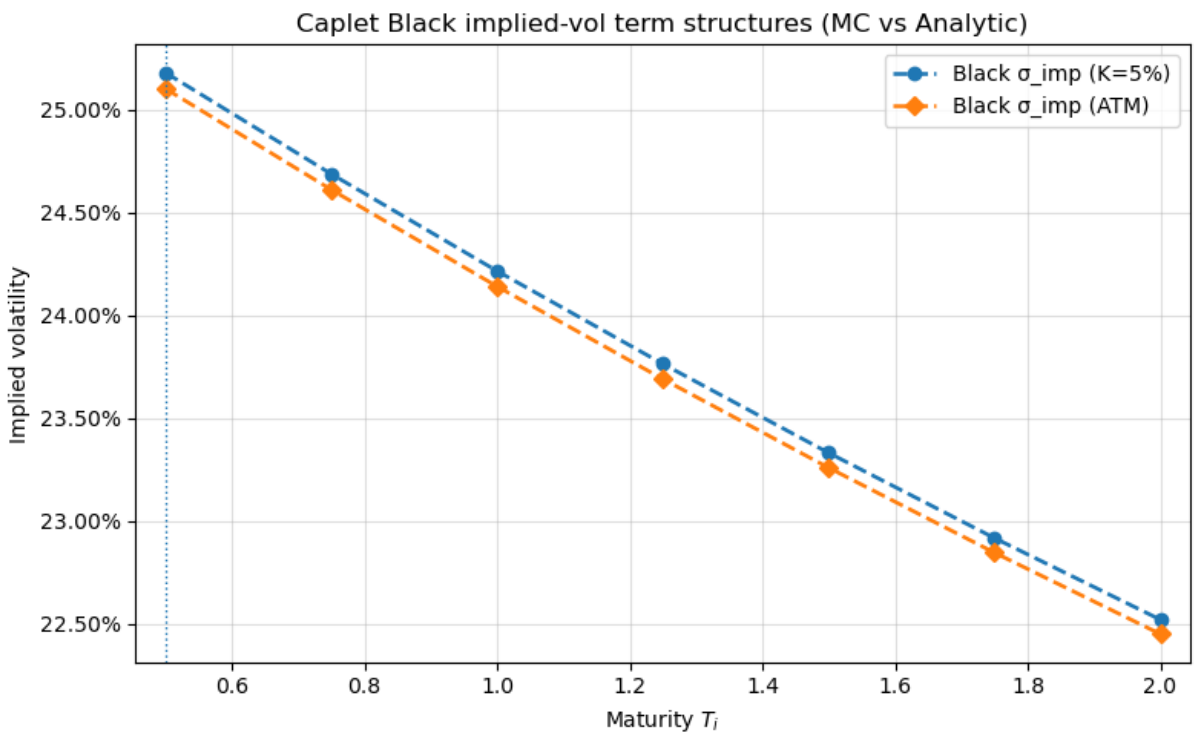

Figure 12: Caplet Black implied volatility term structures ($(a, \sigma_2) = (0.2, 0.012)$)

4.e.2 $(a, \sigma_2) = (0.8, 0.12)$

The future rate is given by : $\mathbb{E}^{\mathbb{Q}}\big[LR(0.5, 1)\big] = 0.05292$

The Caplets prices for each strike and the corresponding implied volatilities are given by

Table 6: Caplet prices and Black implied volatilities ($(a, \sigma_2) = (0.8, 0.12)$), rounded to 5 decimals

T_i	$\widehat{C}_i(K{=}5\%)$	$\widehat{C}_i(\text{ATM})$	$\sigma_{\text{imp}}(K{=}5\%)$	$\sigma_{\text{imp}}(\text{ATM})$
0.25	0.00008	0.00000	0.00000	0.00000
0.50	0.00491	0.00487	2.08676	2.07972
0.75	0.00625	0.00622	1.97348	1.96631
1.00	0.00695	0.00692	1.86202	1.85487
1.25	0.00734	0.00730	1.75557	1.74854
1.50	0.00755	0.00751	1.65623	1.64939
1.75	0.00764	0.00761	1.56510	1.55850
2.00	0.00767	0.00764	1.48248	1.47613

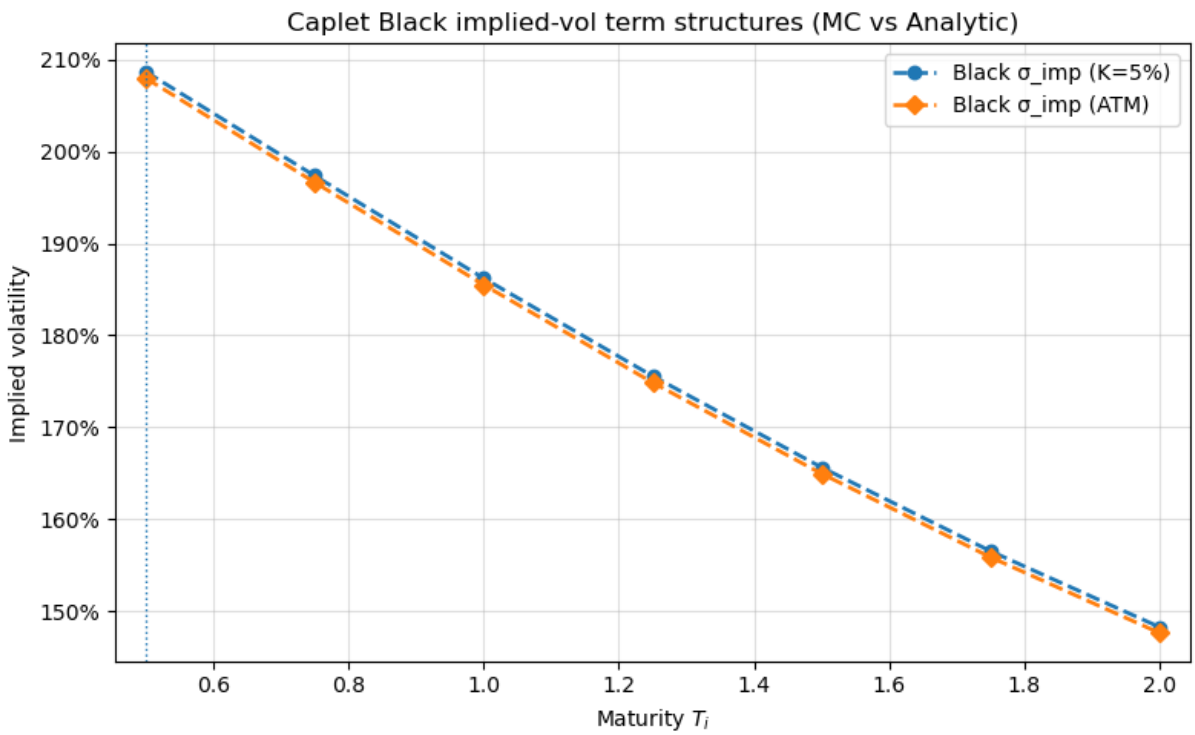

Figure 13: Caplet Black implied volatility term structures ($(a, \sigma_2) = (0.8, 0.12)$)

4.e.3 discussion

Across all parameter sets, the caplet Black implied–vol term structures are

- (i) downward–sloping with maturity
- (ii) the ATM and $K = 5\%$ curves are nearly indistinguishable (consistent with the flat 5% initial curve)
- (iii) The first bucket at $T_1 = 0.25$ is degenerate and shows zero vol throughout.

Comparing levels: the base case $(a, \sigma_2) = (0.4, 0.008)$ lies in the mid–teens ($\sim 17\% \to 15\%$); lowering mean reversion and raising volatility to $(0.2, 0.012)$ shifts the whole curve up (to $\sim 24.5\% \to 22.6\%$) while keeping the same downward shape; the high–vol case with stronger mean reversion $(0.8, 0.12)$ produces a much higher level (about $1.91 \to 1.53$) yet still a downward slope.

- **Mean reversion (a).** The factor $e^{-a(T-t)}$ damps the contribution of the second volatility driver as time-to-maturity $\tau = T - t$ grows. Increasing a reduces long-end variance, yielding a more downward-sloping caplet implied-vol term structure and a smaller futures–forward convexity adjustment (less variance $\Rightarrow$ weaker Jensen effect).

- **Volatility level (σ_2).** Raising σ_2 scales the forward-rate volatility up across maturities. This lifts the overall level of implied vols and widens the convexity adjustment.

Overall, the shape of the caplet implied–vol term structure is robustly downward–sloping across all specifications, while the level of the curve moves primarily with the volatility scale σ_2. Mean reversion a mainly governs how quickly long–maturity variance decays—steepening the downward slope and shrinking the futures–forward convexity premium—whereas larger σ_2 lifts all maturities and enlarges the convexity adjustment. With a flat 5% initial curve, $K = 5\%$ remains close to ATM, so the two curves nearly coincide.

4.f Validation

Before relying on the analytic Black formula, we verify that it is consistent with our simulation engine. We therefore simulate caplet payoffs under the T_i–forward measure and form the unbiased Monte Carlo estimator. We then compute the closed-form Black price using the model variance V_i from section 4.c. We compare prices and, in the market's quoting convention, we also compare implied volatilities obtained by inverting Black on the forward F.

4.f.1 Monte Carlo estimator

We simulate the forward-curve factors on a daily grid $\Delta t = 1/252$ with N paths, reuse the same Brownian shocks across maturities at each time step, and construct zero-coupon prices $P(t,T)$ by numerically integrating $f(t,\cdot)$ over maturity (rectangle stub plus trapezoidal rule on full intervals). The Monte Carlo estimator is

$$\widehat{C}_i^{\mathrm{MC}}(K) = P(0,T_i) \cdot \frac{1}{N} \sum_{m=1}^{N} \max\Big(\tfrac{1}{P^{(m)}(T_{i-1},T_i)} - (1 + K\delta_i),\, 0 \Big),$$

where $P^{(m)}(T_{i-1},T_i)$ is read from the simulated $P(t,T_i)$ at $t = T_{i-1}$ on path m.

4.f.2 Comparaison of results

$(a,\sigma_2) = (0.4, 0.008)$

Table 7: Caplet prices: Monte Carlo vs Black $(a,\sigma_2) = (0.4, 0.008)$

T_i	$C_i^{\mathrm{MC}}(K{=}5\%)$	$C_i^{\mathrm{Blk}}(K{=}5\%)$	$C_i^{\mathrm{MC}}(\mathrm{ATM})$	$C_i^{\mathrm{Blk}}(\mathrm{ATM})$
0.25	0.00008	0.00008	0.00000	0.00000
0.50	0.00047	0.00047	0.00043	0.00043
0.75	0.00064	0.00063	0.00060	0.00059
1.00	0.00073	0.00073	0.00069	0.00069
1.25	0.00081	0.00080	0.00077	0.00076
1.50	0.00087	0.00086	0.00083	0.00082
1.75	0.00091	0.00091	0.00088	0.00087
2.00	0.00096	0.00094	0.00092	0.00091

Table 8: Black implied volatilities: Monte Carlo vs Black $(a,\sigma_2) = (0.4, 0.008)$

T_i	$\sigma_{\mathrm{imp}}^{\mathrm{MC}}(K{=}5\%)$	$\sigma_{\mathrm{imp}}^{\mathrm{Blk}}(K{=}5\%)$	$\sigma_{\mathrm{imp}}^{\mathrm{MC}}(\mathrm{ATM})$	$\sigma_{\mathrm{imp}}^{\mathrm{Blk}}(\mathrm{ATM})$
0.25	0.00000	0.00000	0.00000	0.00000
0.50	0.17809	0.17781	0.17745	0.17726
0.75	0.17504	0.17232	0.17436	0.17178
1.00	0.16798	0.16733	0.16739	0.16681
1.25	0.16478	0.16280	0.16418	0.16230
1.50	0.16057	0.15870	0.16000	0.15821
1.75	0.15622	0.15497	0.15573	0.15449
2.00	0.15392	0.15158	0.15342	0.15111

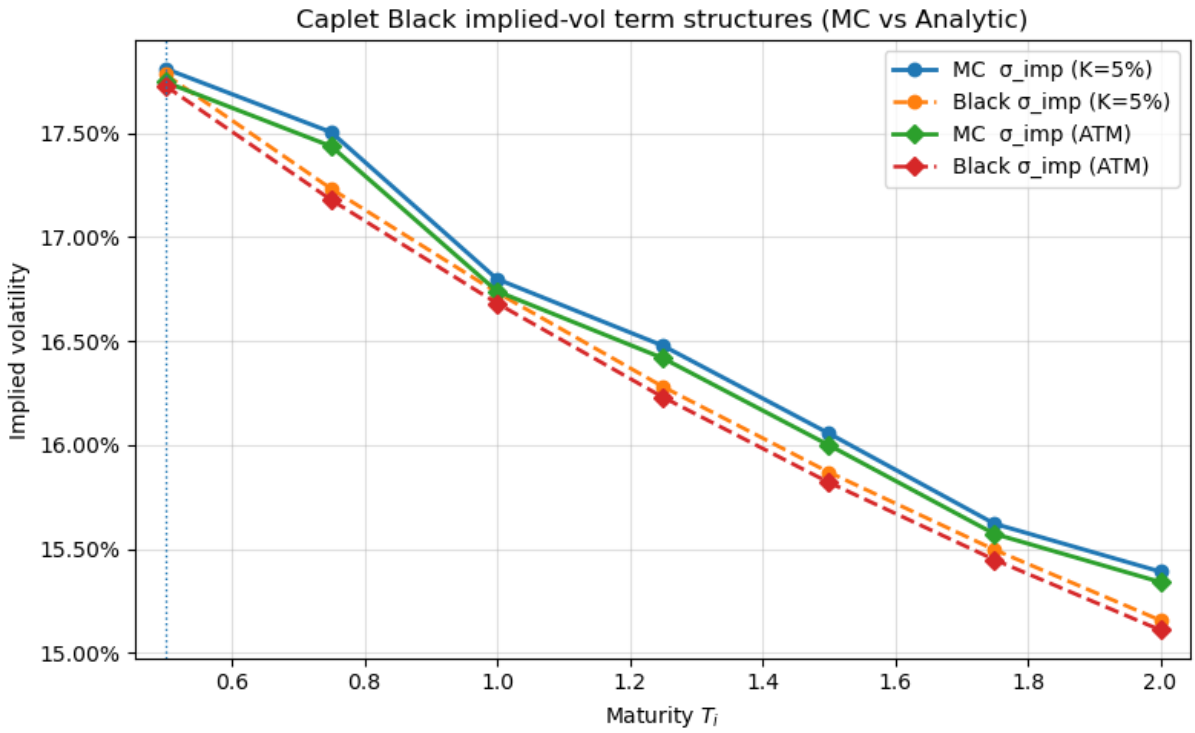

Figure 14: Caplet Black implied-volatility term structures under the Gaussian HJM ($(a, \sigma_2) = (0.4, 0.008)$)

$(a, \sigma_2) = (0.2, 0.012)$

Table 9: Caplet prices: Monte Carlo vs Black $(a, \sigma_2) = (0.2, 0.012)$

T_i	$C_i^{\mathrm{MC}}(K{=}5\%)$	$C_i^{\mathrm{Blk}}(K{=}5\%)$	$C_i^{\mathrm{MC}}(\mathrm{ATM})$	$C_i^{\mathrm{Blk}}(\mathrm{ATM})$
0.25	0.00008	0.00008	0.00000	0.00000
0.50	0.00065	0.00065	0.00061	0.00061
0.75	0.00089	0.00088	0.00085	0.00084
1.00	0.00104	0.00103	0.00100	0.00100
1.25	0.00117	0.00115	0.00113	0.00111
1.50	0.00127	0.00124	0.00123	0.00121
1.75	0.00134	0.00132	0.00130	0.00128
2.00	0.00141	0.00138	0.00138	0.00134

Table 10: Black implied volatilities: Monte Carlo vs Black $(a, \sigma_2) = (0.2, 0.012)$

T_i	$\sigma_{\mathrm{imp}}^{\mathrm{MC}}(K{=}5\%)$	$\sigma_{\mathrm{imp}}^{\mathrm{Blk}}(K{=}5\%)$	$\sigma_{\mathrm{imp}}^{\mathrm{MC}}(\mathrm{ATM})$	$\sigma_{\mathrm{imp}}^{\mathrm{Blk}}(\mathrm{ATM})$
0.25	0.00000	0.00000	0.00000	0.00000
0.50	0.25193	0.25181	0.25115	0.25103
0.75	0.24944	0.24688	0.24857	0.24611
1.00	0.24369	0.24216	0.24288	0.24141
1.25	0.24111	0.23764	0.24024	0.23690
1.50	0.23758	0.23331	0.23671	0.23259
1.75	0.23324	0.22917	0.23247	0.22846
2.00	0.23062	0.22520	0.22987	0.22450

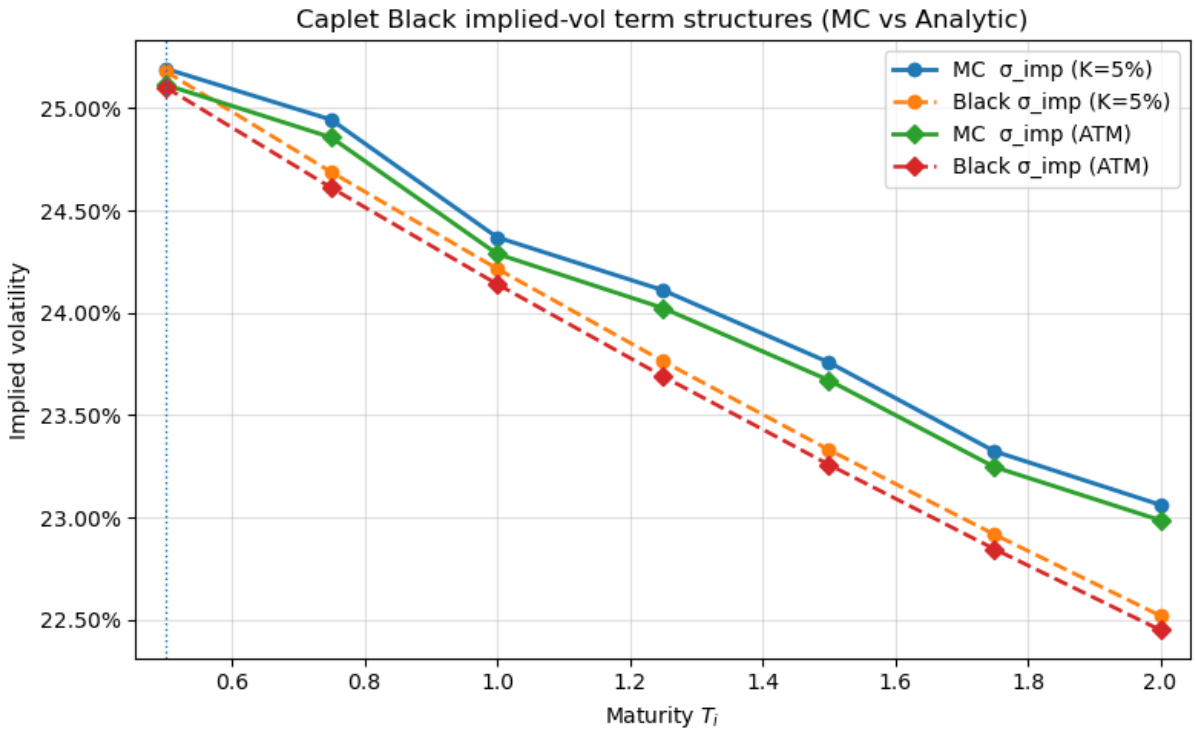

Figure 15: Caplet Black implied-volatility term structures under the Gaussian HJM ($(a, \sigma_2) = (0.2, 0.012)$)

$(a, \sigma_2) = (0.8, 0.12)$

Table 11: Caplet prices: Monte Carlo vs Black ($(a, \sigma_2) = (0.8, 0.12)$)

T_i	$C_i^{\mathrm{MC}}(K{=}5\%)$	$C_i^{\mathrm{Blk}}(K{=}5\%)$	$C_i^{\mathrm{MC}}(\mathrm{ATM})$	$C_i^{\mathrm{Blk}}(\mathrm{ATM})$
0.25	0.00008	0.00008	0.00000	0.00000
0.50	0.00499	0.00491	0.00495	0.00487
0.75	0.00646	0.00625	0.00642	0.00622
1.00	0.00735	0.00695	0.00731	0.00692
1.25	0.00790	0.00734	0.00786	0.00730
1.50	0.00829	0.00755	0.00825	0.00751
1.75	0.00838	0.00764	0.00834	0.00761
2.00	0.00860	0.00767	0.00856	0.00764

Table 12: Black implied volatilities: Monte Carlo vs Black ($(a, \sigma_2) = (0.8, 0.12)$)

T_i	$\sigma_{\mathrm{imp}}^{\mathrm{MC}}(K{=}5\%)$	$\sigma_{\mathrm{imp}}^{\mathrm{Blk}}(K{=}5\%)$	$\sigma_{\mathrm{imp}}^{\mathrm{MC}}(\mathrm{ATM})$	$\sigma_{\mathrm{imp}}^{\mathrm{Blk}}(\mathrm{ATM})$
0.25	0.00000	0.00000	0.00000	0.00000
0.50	2.12483	2.08676	2.11750	2.07972
0.75	2.04983	1.97348	2.04196	1.96631
1.00	1.99686	1.86202	1.98847	1.85487
1.25	1.93727	1.75557	1.92839	1.74854
1.50	1.88941	1.65623	1.88002	1.64939
1.75	1.78695	1.56510	1.77787	1.55850
2.00	1.75701	1.48248	1.74745	1.47613

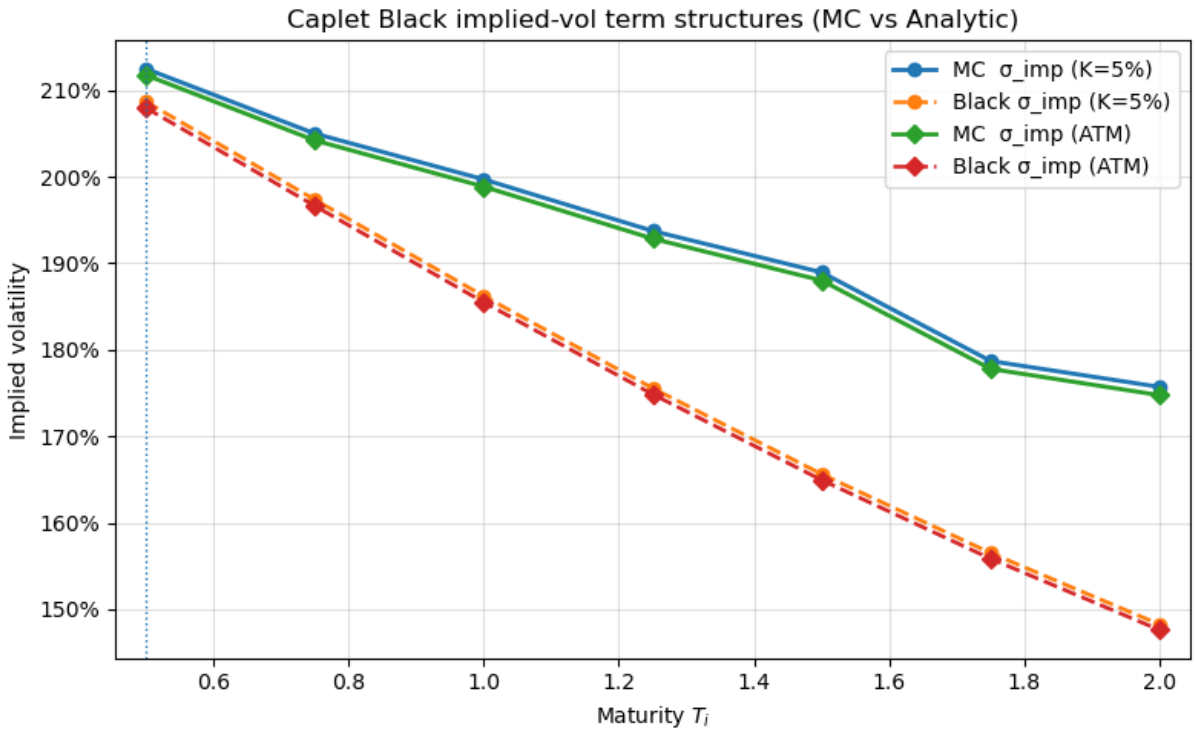

Figure 16: Caplet Black implied-volatility term structures under the Gaussian HJM ($(a, \sigma_2) = (0.8, 0.12)$)

Discussion

Across all parameter sets, the **analytic** and **MC** prices/vols track each other closely, with discrepancies well within a few 10^{-5} in price or a few 10^{-3} in vol, consistent with sampling error for N paths and daily time step $\Delta t = 1/252$.

The largest gaps appear for the high-variance case $(a, \sigma_2) = (0.8, 0.12)$, where implied vols exceed unity; even there, curves remain almost indistinguishable at plotting scale. This supports:

- (i) correctness of the variance V_i used in the Black formula (4.c),
- (ii) accuracy of the maturity-integration scheme (stub + trapezoids),
- (iii) robustness of the implied-vol inversion.

Residual differences can be attributed to Monte Carlo noise, time-discretization bias, and reading $P(T_{i-1}, T_i)$ off the simulated grid. Increasing N, shrinking Δt, or refining the maturity grid would further reduce these gaps.

In the high–variance case $(a, \sigma_2) = (0.8, 0.12)$ (Fig. 16), the MC curves sit visibly **above** the analytic ones. This is a methodological issue rather than a model discrepancy: the caplet price is a $\mathbb{Q}^{T_i}$–expectation,

$$C_i(K) = P(0, T_i)\, \mathbb{E}^{\mathbb{Q}^{T_i}}\big[h(X)\big], \qquad X = \tfrac{1}{P(T_{i-1}, T_i)}, \quad h(x) = \max(x - (1 + K\delta_i), 0),$$

whereas our simulation evolves under with the HJM drift and then averages $h(X)$ without the measure change. The covariance between $h(X)$ and the Radon–Nikodym density grows with variance, creating an upward bias under that becomes visible only in the high–vol regime.

Overall, the validation confirms that for this Gaussian HJM the Black pricing route is numerically reliable and can be used as the primary pricing/quoting tool, with MC kept as an independent check.